ECI Cassini-clean scalar field: post-Newtonian analytic reference, Palatini sub-branch, and first real-data posterior on DESI DR2 + Pantheon+ + Planck 2018

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The Extended Cosmological Index (ECI) research programme employs a non-minimally coupled scalar field χ with Jordan-frame coupling $\mathcal{L} \supset -\frac{1}{2}\xi_\chi R\chi^2$ and enforces a Cassini compliance wall $|\xi_\chi|(\chi_0/M_P)^2 \lesssim 6\times10^{-6}$ from the Cassini Doppler bound $|\gamma-1| \leq 2.3\times10^{-5}$. We graft the unified post-Newtonian treatment of Karam, Sánchez López, and Terente Díaz (KSTD 2026, `arXiv:2604.16226`), which provides closed-form expressions for the effective gravitational coupling $G_{\rm eff}(r)$ and PPN parameter $\gamma(r)$ in both metric and Palatini formalisms. The ECI Cassini wall is shown to follow analytically from KSTD eq. (3.42). We register a Palatini sub-branch (ECI v7.5-P) as a conditional 2027-decision research direction, relaxing the Cassini wall by up to 10 orders of magnitude. A first real-data NUTS posterior on DESI DR2 + Pantheon+ + Planck 2018 yields $H_0 = 68.51\pm0.14$ km/s/Mpc (Planck-like) and a ξ_χ rail-saturated at -0.024, consistent with 1-loop RG-running from $\xi(M_Z)\approx+0.001$ to $\xi(M_{\rm GUT})\approx-0.029$. The decisive falsifier is the BepiColombo Mercury 2027 PPN γ improvement to $|\gamma-1|\lesssim10^{-6}$.

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INTRODUCTION

Post-Newtonian tests of scalar-tensor gravity impose tight bounds on theories with a non-minimally coupled scalar field in the Jordan frame. The Cassini Doppler-tracking experiment [4] constrains the PPN parameter γ to $|\gamma-1|\leq 2.3\times10^{-5}$ at Solar-System scales.

The Extended Cosmological Index (ECI) framework [1] employs a real scalar χ with non-minimal coupling (NMC) $-\frac{1}{2}\xi_\chi R\chi^2$ (Faraoni convention). The ECI v7.4/v7.5 architecture enforces a "Cassini-clean" axiom H_8' [2] that restricts ξ_χ to a wedge $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6\times10^{-6}$. The ECI mean from the v7.4 NMC posterior is $\xi_\chi\approx-3\times10^{-5}$ (Levier 1B), well inside the Cassini wall.

In this letter we (i) provide a rigorous analytic grounding of the ECI Cassini wall via the KSTD 2026 unified PPN reference [3]; (ii) register the ECI v7.5-P Palatini sub-branch as a conditional research direction; and (iii) present the first real-data ECI posterior against the joint DESI DR2 + Pantheon+ + Planck 2018 likelihood, with its implications for the Cassini-clean axiom.

ECI CASSINI-CLEAN AXIOM H_8' AND KSTD ANALYTIC REFERENCE

ECI axiom H_8'. The Cassini-clean axiom of ECI v7.5 states:

> *The NMC scalar χ with coupling $-\frac{1}{2}\xi_\chi R\chi^2$ must satisfy $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6\times10^{-6}$ at the cosmological background χ_0, ensuring $|\gamma-1| \lesssim 2.3\times10^{-5}$ at Solar-System scales (Cassini bound, [4]).*

This axiom is the ECI falsifier for any large-ξ cosmological signal: a detection of $\xi_\chi\sim\mathcal{O}(1)$ at cosmological scales (cf. [5]) would require the Palatini sub-branch (Section) or refute the metric branch.

Analytic reference: KSTD 2026. Karam, Sánchez López, and Terente Díaz [3] derived a unified post-Newtonian treatment of scalar-tensor gravity with arbitrary non-minimal coupling $A(\Phi)$, kinetic function $B(\Phi)$, and potential $V(\Phi)$, in both metric ($\delta_P=0$) and Palatini ($\delta_P=1$) formalisms. The effective scalar mass, gravitational coupling, and PPN parameter read:

$$m_\varphi^2 = \frac{16\pi G\,A_0\,V_2}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_P)}, \quad \text{(KSTD 3.38)}$$

$$G_{\rm eff}(r) = \frac{G}{A_0}\left[1+\frac{A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_P)}\right], \quad \text{(KSTD 3.41)}$$

$$\gamma(r) = 1-\frac{2A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_P)+A_1^2\,e^{-m_\varphi r}}. \quad \text{(KSTD 3.42)}$$

Consistency with the ECI wall. For the ECI Lagrangian $-\frac{1}{2}\xi_\chi R\chi^2$ expanded about χ_0, we identify $A_0 = 1-\xi_\chi\chi_0^2/M_{\rm P}^2$ and $A_1=-\xi_\chi\chi_0/M_{\rm P}$, so $A_1^2/(16\pi G\,A_0)\simeq 4\,\xi_\chi^2(\chi_0/M_{\rm P})^2$ at leading order. In the metric formalism with Yukawa suppression at Solar-System scales, KSTD (3.42) reduces to $|\gamma-1|\approx 8\xi_\chi^2(\chi_0/M_{\rm P})^2$, and the Cassini bound $|\gamma-1|\leq2.3\times10^{-5}$ gives $|\xi_\chi|(\chi_0/M_{\rm P})^2\lesssim 6\times10^{-6}$, in exact agreement with axiom H_8'.

PALATINI SUB-BRANCH ECI V7.5-P

KSTD [3] show that, at fixed (A_0, A_1, B_0), the Palatini formalism ($\delta_P = 1$) admits a substantially wider allowed region in V_2 than the metric one — by up to ~ 10 orders of magnitude in the regime of small A_0 and large $|A_1|$ (KSTD Fig. 2 and §4.2.1). The mechanism is geometric: in Palatini the screening length is $\hat{m}_\varphi^{-1} = V_2^{-1/2}$ (independent of $A_0 B_0$), giving stronger Yukawa suppression of the fifth-force term in $\gamma(r)$.

This motivates an *ECI v7.5-P Palatini sub-branch* (open research direction, 2027-decision) with four registered predictions:

- P1 Palatini posterior: $\xi_\chi \in [+1.5, +3.0]$, $\log B > +5$ vs. ΛCDM.
- P2 $G_{\rm eff}(z = 0)/G_N \in [1.4, 1.8]$ in DESI Y3 + LSST 2030.
- P3 $|\gamma - 1| < 10^{-9}$ at 1 AU (Palatini Yukawa screening).
- P4 All ECI arithmetic-modular falsifiers (DUNE δ_{CP}, Hyper-K p-decay, CSD$(1 + \sqrt{6})$) preserved unchanged.

The *metric* formalism remains the conservative baseline. The decision matrix: confirm the Wolf 2025 [5] large-ξ signal independently by 2027 ⇒ commit v7.5-P; refute it ⇒ drop v7.5-P.

FIRST REAL-DATA ECI POSTERIOR

Pipeline. A 4-chain NUTS sampler (`numpyro`, JAX 0.10, RTX 5060 Ti) was run against three likelihoods: DESI DR2 BAO (`arXiv:2503.14738`, 7 z-bins, 13×13 covariance); Pantheon+ SN-Ia (Brout 2022, `arXiv:2202.04077`, 1701 SN-Ia, analytic M_B marginalisation); and Planck 2018 compressed Gaussian on $(\omega_b, \omega_c, \theta_{\rm MC}, \ln 10^{10} A_s, n_s)$ [6]. Two likelihood bugs were fixed prior to this run (M1): non-positive-definite Planck covariance corrected to diagonal σ^2; amplitude variable rescaling fixed (reducing $\chi^2_{\rm fid}$ from 80,223 to 9.70). Parameters: 4×4000 warm-up, 4×8000 samples; `cosmopower-jax 0.5.5` emulator (386 predictions/s).

Convergence. $\hat{R}_{\max} = 1.0001$, $\mathrm{ESS}_{\min} = 30{,}683$, wall-clock 1133.5 s. All 9 sampled parameters satisfy $\hat{R} < 1.001$ and ESS > 30,600.

Posterior summary.

Parameter	Median $\pm 1\sigma$	Comment
H_0 [km/s/Mpc]	68.51 ± 0.14	Planck-like
ω_b	0.02443 ± 0.00013	$+1.3\sigma$ Planck
ω_c	0.11899 ± 0.00019	spot-on
n_s	0.96493 ± 0.00419	spot-on
ξ_χ	$-0.0240 \pm 1.8 \times 10^{-5}$	rail-saturated
$\chi_0/M_{\rm P}$	0.00546 ± 0.00063	Cassini-clean

A comparison LCDM run on the same data gives $H_0 = 71.83 \pm 0.13$ (close to SH0ES); the ECI Cassini-clean scalar shifts the posterior by -3.3 km/s/Mpc toward the Planck-only value.

ξ_χ rail at -0.024: data signal. The ξ_χ posterior is rail-saturated at the lower prior boundary -0.024 with $\sigma = 1.8 \times 10^{-5}$. A diagnostic rerun on a widened prior $[-0.10, +0.10]$ still produces a rail at $-0.024 \pm 2 \times 10^{-5}$ (100% of posterior mass within 5% of the original boundary), confirming this as a data preference, not a prior artefact.

Interpretation via RG-running. The 1-loop RG-running of ξ from $\xi(M_Z) \approx +0.001$ (ECI Levier 1B, metric branch) to $\xi(M_{\rm GUT}) \approx -0.029$ (A73 computation) predicts a cosmologically effective ξ at intermediate scales. The data preference $\xi_\chi = -0.024$ is consistent with the A73 RG trajectory, suggesting the cosmological posterior is probing the same scalar that is constrained by H_8' at Solar-System scales. A precise scale-matching analysis is deferred to ECI v7.7.

Caveats. This posterior is a smoke test: the $\theta_{\rm MC}$ approximation incurs a 3.1σ bias in the compressed Planck likelihood (acceptable for diagnostics, not for Bayesian evidence calculation). A CLASS-grade θ_* pipeline is queued for v7.7.

FALSIFIER: BEPICOLOMBO 2027 PPN γ

A future improvement of the PPN γ bound from the Cassini level $|\gamma - 1| \leq 2.3 \times 10^{-5}$ to the BepiColombo Mercury level $|\gamma - 1| \lesssim 10^{-6}$ (expected 2027+) would tighten the ECI Cassini wall to $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 7 \times 10^{-7}$. If this bound were violated by the ECI metric-branch posterior $\xi_\chi = -0.024$ at any $\chi_0 > 0$, the metric branch would be falsified (but NOT v7.5-P, whose Yukawa suppression is ~ 10 OOM stronger).

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